# Energy balance in ovariectomized rats with and without estrogen replacement

MARK L. LAUDENSLAGER, CHARLES W. WILKINSON, HARRY J. CARLISLE, AND HAROLD T. HAMMEL
*Department of Psychology, University of California, Santa Barbara, 93106; and Physiological Research Laboratory, Scripps Institution of Oceanography, University of California, La Jolla, California 92093*

LAUDENSLAGER, MARK L., CHARLES W. WILKINSON, HARRY J. CARLISLE, AND HAROLD T. HAMMEL. *Energy balance in ovariectomized rats with and without estrogen replacement.* Am. J. Physiol. 238 (Regulatory Integrative Comp. Physiol. 7): R400–R405, 1980.—The effect of estrogen replacement on several parameters of energy balance was investigated in ovariectomized rats tested during the dark phase of their diurnal cycle. Estrogen replacement, either as 17$\beta$-estradiol or $\beta$-estradiol-3-benzoate via subcutaneous Silastic capsules, was associated with elevated rates of heat production and dry heat loss relative to untreated ovariectomized controls. Estrogen treatment reduced body mass and retarded fur growth. The effects of estrogen replacement on heat production and dry heat loss could not be attributed to these differences in body mass and fur growth or locomotor activity. Estrogen replacement had no effect on rate of evaporative heat loss. If estrogen replacement was delayed 75 days following ovariectomy, the increase in heat production and dry heat loss was not observed. There was no effect of the hormone treatment on rectal temperature. It was concluded that either heat production was elevated, with dry heat loss increased to compensate for the additional thermal load, or dry heat loss was accelerated with heat production elevated in compensation.

metabolic heat production; oxygen consumption; dry heat loss; gradient layer calorimetry; evaporative heat loss; rectal temperature

CYCLIC VARIATIONS IN BODY TEMPERATURE associated with the ovarian cycle have been observed frequently (3, 16, 18, 19). There seems to be general agreement that body temperature is elevated when progesterone levels are high, as in the luteal phase of the human menstrual cycle (18) or during diestrus, pregnancy, or pseudopregnancy in the rat (3, 19). Furthermore, exogenous administration of progesterone is associated with a rise in body temperature (8, 14, 16, 19). Conversely, the role of estrogens is not as well defined. Exogenous administration of estrogens has been associated with both increased (16) and reduced (11, 14, 20, 24) body temperature. Both high (16, 26) and low (3, 19, 26) body temperatures have been observed at proestrus-estrus, following the period of high estrogen secretion during the morning of proestrus (12). The exact timing of these measurements may be critical, but based on body temperature measurements alone, the role of estrogens in these changes is far from established.

Another approach toward understanding the role of estrogens in thermal balance is to measure thermoregulatory responses in addition to body temperature following exogenous administration of estrogens. Basal metabolic rate is depressed in humans following ovariectomy, but it can be restored with supplemental administration of estrogens (6). Behavioral thermoregulatory heat-acquisition responses are higher in estrogen-treated ovariectomized rats relative to untreated ovariectomized rats (24). Paradoxically, rectal temperature of the estrogen-treated rats was lower than untreated rats at the end of the behavioral tests. This observation suggests that perhaps metabolic heat production is reduced and/or rate of heat loss is increased in estrogen-treated rats. An apparent confirmation of an elevation in rate of heat loss has been reported for ethinyl estradiol-treated rats which cooled faster than untreated rats when cold stressed; it was also noted that the rates of oxygen consumption (heat production) for the two groups were not different (9). The present study further examines the effects of sustained estrogen replacement in ovariectomized rats on several parameters of energy balance.

## MATERIALS AND METHODS

*Experiment 1: effect of estradiol on heat production and dry heat loss.* (*Experiment 1* was performed at Scripps Institution of Oceanography, University of California, while M. L. Laudenslager was a postdoctoral fellow.) Six adult female Sprague-Dawley rats, weighing 193–208 g at the time of ovariectomy, were caged individually in a chamber maintained at 22 ± 2°C. Purina lab chow and water were freely available except for a 12- to 18-h food-deprivation period to all tests. The rats were kept on a reversed day-night cycle of LD 12:12 and tested at approximately the same time during the dark portion of the cycle.

The experimental apparatus consisted of a 15.2 x 15.2 x 15.2 cm (ID) gradient layer calorimeter as previously described (5). The rat was placed in a 10.5 x 9 x 14 cm screen cage inside the calorimeter, which was maintained within ±0.05°C of any desired temperature. Air, dried to a relative humidity of less than 1% by passing it through concentrated $H_2SO_4$ and Drierite, was drawn at a constant rate (range, 1.8–2.2 l/min) through the calorimeter.


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A portion of the effluent air was redried and passed through a Beckman F-3 paramagnetic oxygen analyzer, recombined with the remaining effluent air, and passed to a dry gas volume meter. The oxygen analyzer was calibrated by a partial pressure method (5). Ambient temperature ($T_a$) and mean calorimeter wall temperature were measured with 36-gauge copper-constantan thermocouples referenced to a melting ice bath. The voltage outputs from the calorimeter, oxygen analyzer, and thermocouples were recorded on a 12-point recording potentiometer (Leeds and Northrup, North Wales, PA). Rate of heat production was determined from the average rate of oxygen consumption (STPD) assuming a respiratory quotient of 0.80 and 10 ml of $O_2$ consumed/min as equivalent to 3.38 W. Rate of dry heat loss was determined from the average output of the gradient-layer calorimeter.

The rats were ovariectomized at approximately 60 days of age. Two weeks after ovariectomy, they were randomly assigned to one of two groups. *Group A* received a single subcutaneous, cervical Silastic capsule implant (5 mm long; 3.2 mm OD; 1.6 mm ID) containing 17$\beta$-estradiol ($E_2$) (Sigma Chemical, St. Louis, MO), and *group B*, the controls, received an identical but empty capsule. The ends of the capsules were sealed with wood applicator sticks so that diffusion was only through a 5-mm length of tubing between the sticks. Silastic capsules such as these produce a sustained release of $E_2$, which is dependent on the length and wall thickness of the tubing (21). Testing began 1 wk later. The rats were tested individually in the dark for 2 h at 2.5, 10, 20, and 30°C. Rectal temperature ($T_{re}$) was measured with a telethermometer (Yellow Springs Instrument, Yellow Springs, OH; model 46 TUC) before and after each test with the vinyl probe (no. 402) inserted 6 cm. Tests were separated by at least 96 h. Only one $T_a$ was presented on any day, and each rat was tested twice at each $T_a$. The average rates of heat production and dry heat loss were determined from the last 90 min of the test. The hormone treatments were reversed 75 days following ovariectomy, and the rats were retested 10 days later as described previously.

*Experiment 2: comparison of the effects of estradiol and estradiol benzoate on heat production and evaporative heat loss under conditions of restricted activity.* The second experiment (performed at the Department of Psychology, University of California, Santa Barbara) replicates the general observations of *experiment 1* with regard to heat production in addition to comparing the relative efficacies of 17$\beta$-estradiol and $\beta$-estradiol-3-benzoate because estrogen frequently has been administered as the longer acting ester (4, 16, 22, 24). Second, the influence of restricted activity on the observed increase in heat production is also examined, since increased activity has been associated with high endogenous estrogen levels (3, 23, 27) or estrogen injections (22).

Twelve adult female Sprague-Dawley rats, weighing 151–214 g at the time of ovariectomy, were maintained in an animal colony room kept at 24 ± 1°C. Otherwise, maintenance was identical to *experiment 1*.

The experimental apparatus consisted of a 21 x 12 x 11 cm (ID) Plexiglas chamber through which dried air was drawn at a constant rate (range, 1.6–2.1 l/min). This chamber was placed in an environmental chamber maintained within ±0.1°C of the temperatures specified in RESULTS. The Plexiglas chamber was fitted with a hardware cloth floor, under which a 1-cm layer of mineral oil trapped urine and feces. The effluent air was passed over a wide-range humidity sensor (no. 15-1811, Hygrodynamics, Silver Spring, MD) enclosed in a 1.0-liter paint can. The sensor output was monitored with a Hygrodynamics Digital 1 electronic hygrometer. A portion of the air was redried and directed to a Beckman OM-11 polarographic oxygen analyzer, the output of which was measured on a recording potentiometer (Leeds and Northrup). Total volume flow was measured with a dry gas volume meter. The accuracy of the system was checked by burning ethanol in the animal test chamber (5). The measured $O_2$ consumption agreed with the predicted $O_2$ consumption within ±0.5%. Evaporative water loss (mg/min) for $T_a$ of 25 and 15°C was determined by multiplying the average relative humidity by the flow and the density of saturated steam for that $T_a$, and this was converted to W/kg. Condensation of vapor in the air lines prevented accurate measurement of evaporative water loss for tests at 5°C.

Following ovariectomy at 50–60 days of age, the rats were randomly divided into three equal groups. Earlier observations (24) indicated that capsules 5 mm in length might produce relatively high plasma $E_2$ levels. Consequently the length of the capsules between the wood sticks was reduced to 2 mm for *experiment 2*. *Group I* received one cervical subcutaneous Silastic capsule implant (2 mm long; 3.2 mm OD; 1.6 mm ID) containing 17$\beta$ estradiol (Sigma Chemical); *group II* received the same size implant but containing $\beta$-estradiol-3-benzoate (EB) (Sigma Chemical); *group III* received empty implants (C) as a control procedure. At least 1 wk was allowed before testing began. Each rat was tested in the dark twice in a random order, once unrestrained in the large Plexiglas chamber and once in a small 14 x 5.5 x 5.5 cm hardware cloth cage placed inside the larger Plexiglas cage. The size of the small cage was such that locomotor activity by the rat was severely restricted. Each rat was adapted for 3–4 h to the small cage 2–3 days prior to testing. During the first 2 h of a test, $T_a$ was maintained at 25°C and then lowered over a 20- to 25-min interval to 15°C and 2 h later reduced to 5°C for 2 h. Rectal temperature was measured before and after each test. Only the last 60 min of exposure to any $T_a$ was used for determining average heat production and evaporative heat loss.

*Radioimmunoassay.* At the conclusion of *experiments 1* and *2*, the rats were anesthetized with ether, and blood samples were collected by cardiac puncture. Samples were centrifuged, and the plasma was removed and frozen until assayed. Plasma $E_2$ levels were determined by radioimmunoassay, with a clinical procedure (1) modified by L. D. Keith (unpublished) for low $E_2$ concentrations. Briefly, $E_2$ was ether extracted from a 0.5-ml plasma sample and purified by column chromatography. The antiserum (S-52 no. 5) was obtained from Dr. G. Abraham (Harbor General Hospital, Torrance, CA). Hexalabeled [2,4,6,7,16,17-$^3$H(N)]estradiol, sp act, 135 Ci/mM, (NET-517, New England Nuclear, Boston, MA) served

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as tracer. The antiserum and extracted sample were mixed and allowed to equilibrate at room temperature for 2 h. This solution was cooled to 0°C in an ice bath at which time the tracer was added. The mixture of antiserum, extracted sample, and tracer was allowed to incubate overnight at 5°C. The reaction was stopped with dextran-coated charcoal. The samples were centrifuged, and the entire supernatant was added to the liquid scintillation solution (8 g butyl PBD/1,000 ml toluene). The samples were counted for 10 min each on a Beckman liquid scintillation counter. Minimal detectable levels of $E_2$ using this assay, based on standard curves run with each group of samples, ranged from 1 to 2 pg. Statistically significant displacement of tracer was produced by 8–12 pg $E_2$/ml plasma after correction for procedural losses and solvent effects.

*Statistical analysis.* The effects of hormone treatment, $T_a$, order of hormone administration (*expt 1*), and cage size (*expt 2*) on heat production, dry heat loss (*expt 1*), evaporative heat loss (*expt 2*), and $T_{re}$ were tested by analyses of variance for repeated measures. Where appropriate, *t* tests for differences between two means were used.

## RESULTS

*Experiment 1.* Ovariectomized rats receiving estrogen replacement showed a marked increase in rates of heat production and dry heat loss relative to rats receiving empty capsules. Regardless of $T_a$, rates of heat production ($P < 0.001$) and dry heat loss ($P < 0.005$) were elevated in the $E_2$-treated rats. Average rates of heat production and dry heat loss noted for both groups, before and following hormone treatment reversal, are shown in Fig. 1. For both groups, rates of heat production ($P < 0.001$) and dry heat loss ($P < 0.001$) increased as $T_a$ decreased. *Group A* showed a substantial drop in rates of heat production and dry heat loss when the $E_2$ treatment was discontinued, but *group B* showed little change in heat production and dry heat loss when $E_2$ was restored 75 days following ovariectomy. Yet after treatment reversal, the $E_2$-treated rats (*group B*) still maintained the higher rates of heat production and dry heat loss relative to the untreated rats (*group A*). Thus, a significant treatment order × hormone interaction was noted for heat production ($P < 0.001$) and dry heat loss ($P < 0.001$). For any $T_a$, rates of heat production ($P < 0.01$) and dry heat loss ($P < 0.005$) were consistently lower the second time the rats were tested relative to the first test.

The hormone treatment had no main effect on either pretest or posttest $T_{re}$. However following treatment reversal, *groups A* and *B* both showed increases in pre- and posttest $T_{re}$ resulting in a significant treatment order × hormone interaction ($P < 0.01$). Average pre- and posttest $T_{re}$ observed for both groups during both hormone treatments are given in Table 1. Posttest $T_{re}$ at $T_a$ of 2.5 and 10°C were higher relative to posttest $T_{re}$ at 20°C. Posttest $T_{re}$ was also higher for tests at 30°C relative to 20°C. Thus, $T_a$ had a significant effect on posttest $T_{re}$ ($P < 0.001$).

Body masses of $E_2$-treated rats were lower than nontreated rats ($P < 0.001$) as shown in Fig. 2. Prior to

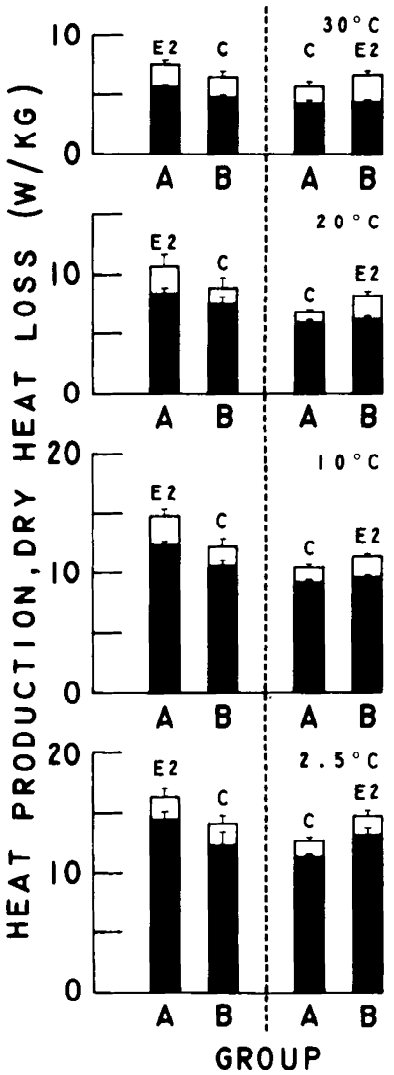



FIG. 1. Heat production (*unshaded portion*) and dry heat loss (*shaded portion*) as a function of $T_a$ (indicated in upper right-hand corner of each panel) and hormone treatment before and following treatment reversal. *Bracket* indicates 1 SEM. Estrogen replacement is indicated by $E_2$ and untreated control condition is indicated by a C.

TABLE 1. *Pretest and posttest $T_{re}$ (°C) for both groups for all hormone conditions*

| Group | Hormone Condition | $T_a$, °C | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | 2.5 | | 10 | | 20 | | 30 | |
| | | Pre | Post | Pre | Post | Pre | Post | Pre | Post |
| *A* | $E_2$ | 38.5 ±0.81 | 38.3 ±0.60 | 38.2 ±0.63 | 37.9 ±0.52 | 38.3 ±0.47 | 37.7 ±0.37 | 38.4 ±1.07 | 38.2 ±0.87 |
| | C | 39.1 ±0.45 | 38.7 ±0.33 | 38.9 ±0.48 | 38.6 ±0.36 | 38.8 ±0.63 | 37.6 ±0.23 | 39.0 ±0.71 | 38.5 ±0.26 |
| *B* | $E_2$ | 38.1 ±0.70 | 38.1 ±0.44 | 38.2 ±0.46 | 37.5 ±0.08 | 38.2 ±0.45 | 37.5 ±0.46 | 38.4 ±0.52 | 38.0 ±0.39 |
| | C | 38.4 ±0.64 | 38.7 ±0.16 | 37.5 ±0.58 | 38.4 ±0.39 | 38.1 ±0.43 | 38.2 ±0.45 | 38.3 ±0.63 | 38.7 ±0.48 |

Values are means ± SD.

treatment reversal body masses of *group A* were less than *group B*; when $E_2$ replacement was given to *group B*, their body masses dropped below those of *group A* whereas the body masses of *group A* increased above

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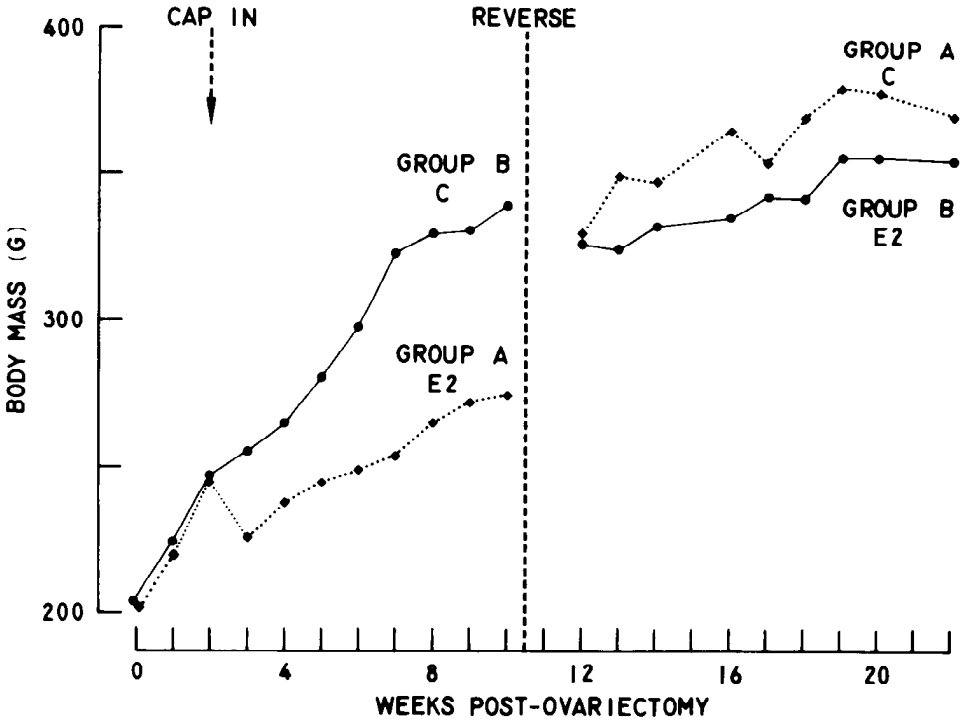



FIG. 2. Growth curves for *groups A* (*diamonds*) and *B* (*circles*) before and following treatment reversal. *Weeks 0* (ovariectomy), *1*, and *2* indicate average nondeprived body masses, whereas all subsequent body masses indicate averaged 18-h food-deprived noted before testing. *Cap in* indicates point at which capsules were implanted in both groups.

their previous level. The difference in body mass was greatest prior to treatment reversal; thus, a significant treatment order × hormone interaction was observed for body mass ($P < 0.001$). Finally, body mass was consistently higher prior to the second test at any $T_a$ relative to the first test ($P < 0.001$).

Irrespective of when $E_2$ was administered, $E_2$-treated rats appeared to have less dense fur than the untreated rats. After a 2–3 $cm^2$ patch was shaved on the neck for the implantation of the capsules, fur growth appeared retarded in that region of $E_2$-treated rats. If the untreated rats had better insulation due to thicker pelage, they might be expected to have reduced heat production, since heat loss would be retarded. If the only effect of $E_2$ is to reduce insulation, removing the fur from both groups should equalize rates of heat loss. After all tests were completed, both groups were shaved completely and then tested twice each at 10 and 20°C. At 10°C, average rates of heat production and dry heat loss, respectively, were 14.90 ± 0.84 (SE) W/kg and 13.31 ± 0.58 for the untreated group and 16.97 ± 0.56 and 14.80 ± 0.35 W/kg for the $E_2$-treated group. At 20°C, the untreated group had a mean rate of heat production and dry heat loss of 9.22 ± 0.54 and 7.91 ± 0.34 W/kg, respectively, whereas the $E_2$-treated group had averages of 11.30 ± 0.21 and 8.87 ± 0.08 W/kg, respectively. Thus, when both groups were equated for possible fur differences, the $E_2$-treated rats still had higher rates of heat production and dry heat loss.

*Experiment 2.* Administration of either $E_2$ or EB was associated with an elevation in rate of heat production relative to that observed in untreated rats, thus replicating the observations of *experiment 1* with regard to heat production. Furthermore, restricting locomotor activity by confining the rats to small cages during testing did not attenuate the increase in heat production. The effects of hormone treatments ($P < 0.05$) and $T_a$ ($P < 0.001$), but not cage size ($P > 0.05$), were significant and are illustrated in Fig. 3. Post-hoc comparisons indicated that rates of heat production were greater than controls for both the $E_2$ ($P < 0.001$) and EB ($P < 0.001$) groups, but they were not different from each other ($P > 0.05$).

Rate of evaporative heat loss was not influenced by hormone treatment ($P > 0.05$) or cage size ($P > 0.05$). Average rates of evaporative heat loss combined across all groups tested in the large cage were 0.61 ± 0.50 (SE) W/kg and 1.24 ± 0.16 W/kg for $T_a$ of 15 and 25°C, respectively. In the small cage, average rates of evaporative heat loss for $T_a$ of 15 and 25°C were 0.42 ± 0.04 and 0.96 ± 0.14 W/kg, respectively. The effect of $T_a$ on rate of evaporative heat loss was significant ($P < 0.001$).

The hormone treatment did not affect pretest or posttest $T_{re}$. The average $T_{re}$ prior to tests was 37.4 ± 0.20 (SD)°C, 37.6 ± 0.18°C, and 37.5 ± 0.16°C for the $E_2$, EB, and control groups, respectively. Posttest $T_{re}$ was 37.5 ± 0.10, 37.5 ± 0.34, and 36.6 ± 0.59°C following small cage tests and 37.7 ± 0.35, 37.7 ± 0.30, and 38.0 ± 0.26°C after unrestricted tests for the $E_2$, EB, and control groups, respectively. Posttest $T_{re}$ was lower following tests in the small cage for all treatment groups ($P < 0.001$). Finally, as observed in *experiment 1*, rats receiving $E_2$ (266 ± 26.0 (SD) g) or EB (252 ± 23.2 g) replacement were smaller in mass than the untreated controls (315 ± 30.3 g) ($P < 0.05$) and also appeared to have poorer fur growth.

*Radioimmunoassay.* The rats from *experiment 1*, which received 5.0-mm Silastic capsules containing $E_2$, had mean plasma estradiol concentrations of 49.4 ± 5.4 (SD) pg/ml, whereas the rats receiving empty capsules had plasma estradiol concentrations below the minimal detectable level of the assay. The rats from *experiment 2*, which received 2-mm Silastic capsules containing $E_2$, had plasma estradiol concentrations of 31.5 ± 8.8 pg/ml;

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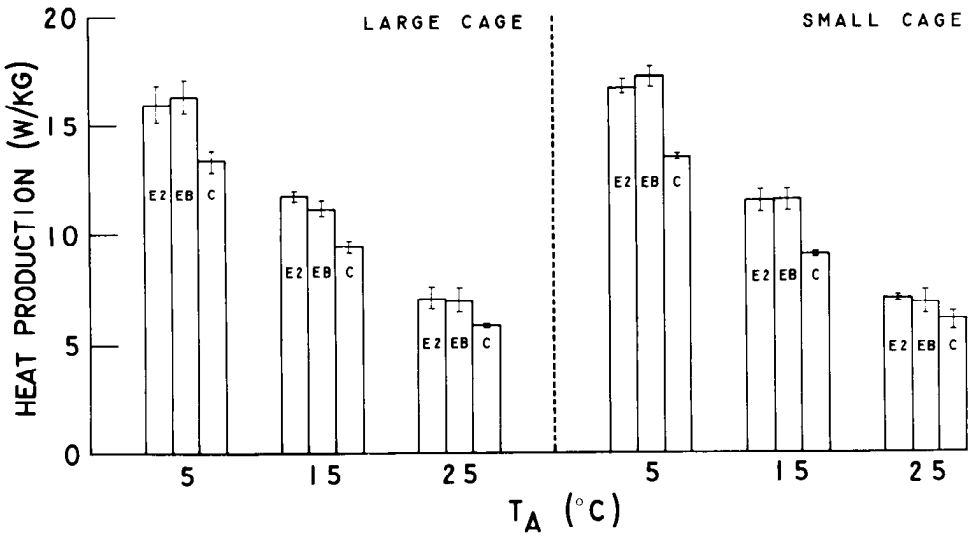



FIG. 3. Heat production as a function of $T_a$, hormone treatment, and cage size for *expt 2*. *Brackets* enclose ±1 SEM.

those receiving 2-mm EB-containing capsules had plasma estradiol levels of 36.6 ± 15.4 pg/ml. The rats from *experiment 2* which received empty capsules also had plasma estradiol levels below the minimal detectable level of the assay.

## DISCUSSION

The present results indicate that nocturnal rates of heat production and dry heat loss are elevated in ovariectomized rats maintained with sustained plasma estradiol levels within the range reported for late diestrus and early proestrus in intact rats (12). The increased behavioral activity typically associated with estrogens (3, 22, 27) might be sufficient to produce the elevated heat production. However, limiting activity as in *experiment 2* did not attenuate the increased heat production. It can be argued that the rats struggled against the restraint, but frequent observation of the rats during tests in the small cages failed to indicate that the rats were struggling or attempting to escape. It seems unlikely that the observed elevation in heat production can be accounted for simply on the basis of increased activity.

Reduced body mass (Fig. 2) is commonly associated with estrogen treatment (23). The familar surface rule (15) predicts that heat production per unit body mass will increase as body mass decreases. Because of the reduced mass of the $E_2$-treated animals, this relation predicts an 8% increase in basal heat production relative to the controls in *experiment 1* prior to treatment reversal when the difference in body mass was greatest. A 4% increase relative to controls is predicted for the estrogen-treated rats of *experiment 2*. In both instances, these predicted increases are insufficient to account for the 14–17% higher rates of heat production noted in both experiments for $T_a$ of 25 and 30°C. The analysis of covariance (25) can adjust for the differences a concomitant variable such as body mass might produce in a dependent variable such as heat production. Treating body mass as a covariate, the hormone effect remains significant for both heat production ($P < 0.001$) and dry heat loss ($P < 0.01$). Interestingly, analysis of covariance revealed that the lower rate of heat production observed during the second test at each $T_a$ was attributable to an increase in body mass occurring in both groups over time (see Fig. 2). However, the reduction in rate of dry heat loss remained significant ($P < 0.05$). Familiarization to the testing conditions may have also contributed to lowering of heat production and dry heat loss in both groups on subsequent tests.

The increase in blood pressure of spontaneously hypertensive female rats can be attenuated by daily estrogen supplements (13), and evidence also exists that estrogen treatments alter cardiovascular responsiveness to vasoactive substances such as isoproterenol (2, 10, 11). Estrogen replacement in the present study may have interfered with the thermoregulatory vasomotor control in such a way as to produce accelerated dry heat loss as has been previously suggested (9). If, under the conditions of the present study, dry heat loss were accelerated as a primary effect of estrogen, perhaps heat production was elevated in compensation. Since the present study is based on steady-state measurements, it cannot be determined if the initial effect of estrogen treatment is to increase heat production or dry heat loss. It is just as likely that the primary effect of estrogen was to directly increase heat production, and dry heat loss has increased either passively or as an active compensation for the elevated heat production. Regardless of the appropriate interpretation, the changes were compensated for in such a way as to maintain thermal balance.

An unexpected observation was the failure of delayed $E_2$ replacement to produce a large increase in heat production and dry heat loss of *group B* (see Fig. 1). Thus, delay of estrogen replacement for 75 days following ovariectomy would appear to have reduced or blocked the effects noted when $E_2$ is administered immediately. Others have noted differences in responsiveness to estrogen following delays in replacement. For example, the effectiveness of estrogen in stimulating lordosis behavior in ovariectomized rats is attenuated when the estrogen treatment is delayed for 6 wk (7). With continued estrogen treatment, normal behavioral responses were observed in 12 days although uterine mass and plasma luteinizing hormone levels never returned to control values. It has also been suggested that estrogen receptors

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degenerate during periods of estrogen withdrawal, but new receptor activity reappears within 6 h following a single estrogen injection (17). Functional changes in estrogen binding of *group B* could account for their failure to elevate heat production when $E_2$ replacement was delayed, but this is yet to be substantiated.

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Present address of C. W. Wilkinson: Dept. of Physiology, School of Medicine, University of California, San Francisco, CA 94143.

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